



A bio-inspired liquid-like smooth copolymer coating with superior biofouling resistance and drag reduction

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ABSTRACT

The marine biofouling and fluid resistance have a detrimental impact on underwater vehicles. It is essential to develop functional material integrated with both antifouling and drag reduction effects for maritime transportation and navigation. Herein, to avoid the disadvantages of common slippery liquid-infused surfaces, such as the complexity of preparation micro/nanopores and the depletion of the lubricant layer, the novel liquid-like coating was fabricated by integrating the synthesized long chain organosilicon monomer (HPSM) to the self-polishing polymer chains. It is that the ingenious design integrated anti-fouling and drag reduction functions. The surface composition and structure, mechanical properties, antifouling and drag reduction properties of the coating were characterized by FT-IR, XPS, SEM, μ -PIV. These results indicated that the unique monomer HPSM endowed the coating with unique liquid-like properties, including special wettability, viscoelasticity and extremely low roughness. The flexibility of HPSM promoted the self-adaption ability of the coating in the seawater environment, which can effectively either reduce the fluid resistance when the fluid flowed through the surface or the adhesion strength of the biofouling, such as lots of protein (BSA) and alga (*Porphyridium* sp. and *Navicula* sp.). Moreover, the synergistic effect between liquid-like properties and the interface renewal behavior of self-polishing components further strengthened the antifouling performance of the coating. The marine field test further indicated the comprehensive anti-fouling performance and mechanical stability of the coating in practical application scenarios. This research provided a facile approach for preparing antifouling and drag-reduction coating. Therefore, it was believed to be a great inspiration for the design of functional coating against biofouling and drag reduction in the ship and marine industry field.

1. Introduction

Biofouling is a phenomenon that the kinds of marine organisms, such as lime worms, algae, larvae, barnacles, and mussels, adhere to and accumulate on the surface of underwater equipment, which has caused enormous economic losses for the exploitation of marine resources and the maritime transport industry of humans [1–3]. For example, when the ships suffer from biofouling, sailing resistance will suddenly increase and fuel consumption will be augmented. Meanwhile, the microorganisms adhering to the hull surface will secrete a mass of metabolites to aggravate hull corrosion and shorten their service time. Moreover, periodic manual removal of a great deal of microorganisms will enlarge the

maintenance cost and berthing time [4–6]. Therefore, various novel antifouling/antibacterial materials and protection technologies have been developed over the past few decades [7,8], and organotin coatings were widely used because of their strong insecticidal effect and easy preparation [9–12]. However, due to metal tin's toxicity and side effects, the International Maritime Organization (IMO) banned the application of coatings with organotin in 2008 [13]. Therefore, antifouling materials with broad-spectrum, efficient antifouling ability and fewer environmental side effects have been intensively researched in recent years [14–16].

A variety of environmentally friendly antifouling surfaces without biocides have been developed, including self-assembly coatings [17,18],

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dynamic degradation surfaces [19], hydrogel coatings [20–22], self-stratified coatings [23], and wettability regulation surfaces [24–26]. Surfaces with unique wetting properties have attracted considerable attention due to their unique liquid-repellency properties in this field. For example, slippery liquid-infused surfaces (SLIPs) have been conducted for widespread research since they were proposed by the Aizenberg group in 2011 [27]. SLIPs are a special kind of solid surface involving lubricant liquid with low polarity and rough solid surface with micro/nano characters, where the capillary attractive force of micro-structure anchors the slimy liquid and forms a stable slippery liquid layer [28]. David M. Lynn's group fabricated an antifouling SLIPs in flexible narrow-bore tubing using the LBL method without compromising their inherent slippery and antifouling behaviors under repeated bending and coiling [18]. In addition, Fu's research group bonded the amino-terminated dimethyl siloxane to the substrate surface. It achieved excellent antifouling and drag reduction properties through modified nanoparticles with solid adhesion to the substrate [28]. Although SLIPs are novel and attractive materials, their specific functions, such as antifouling, drag reduction, antifogging and icing prevention, depend on the distinctive filled lubricant to achieve a strong repulsion for the probe liquid. The filling and probe liquid's physical and chemical properties must satisfy certain constraints, such as surface tension, immiscibility, and others [27,29]. In addition, SLIPs are restricted by the poor retention capacity of the lubricant which can be depleted over time by evaporation [30], lost under shear stresses and external abrasion [31], or reduced due to the cloaking layers and wetting ridges around the probe droplets [32,33]. To address these shortcomings, some alternatives insensitive to depletion induced by evaporation, cloaking or shear forces, such as high viscosity perfluorinated compounds, nonvolatile ionic liquids, low melting point solid paraffin waxes and gelators containing oils, were developed to prepare more durable SLIPs [34]. Furthermore, the organogels and other replenishing slippery liquid surfaces have been fabricated [35–37]. These interesting works effectively reduced the loss of lubricant and extend the antifouling action. However, they did not fundamentally solve the problem. Meanwhile, it is usually time consuming, complicated, and even necessary for sophisticated instruments to create the micro/nanostructures on solid surfaces. These problems seriously restrict the practical broad application of SLIPs interface.

It is one important route to construct slippery surfaces by filling super slippery liquids into porous structures, and it is also a facile approach to learning from nature [38,39]. Nature has many slippery surfaces, such as pitcher plants and dolphin skin. Interestingly, dolphins can prevent themselves from the adhered biofouling during their life and are one of the fastest animals in the ocean [40]. Scientists have proposed many different mechanisms of drag reduction in dolphins, such as the heat released by skin reducing the fluid's viscosity, flexible wall of resistance reduction and the synergy of multiple mechanisms. It is commonly accepted that the smooth and compliant skin is the main factor for the high-speed movement of dolphins [41].

Herein, inspired by the charming properties of the smooth and compliant skin of the dolphins and common SLIPs for antifouling and drag reduction, a linear long-chain polymer (HPSM) with good compliance and lubrication properties was intelligently anchored on the interface with self-polishing properties to prepare a smooth liquid-like surface (SLLS). The moving freely Si-O-Si of the polymer chain was like a layer of liquid, and the SLLS possessed unique surface wettability, viscoelasticity, surface tension and smoothness. Both high drag reduction and ideal antifouling performance were achieved for the coating. This research addressed these questions of the complex preparation process and lubricant loss of lubricant for traditional SLIPs by integrating organosilicon molecular chain with a polymer. An eco-friendly and attractive surface with superior antifouling and drag reduction capability was fabricated by facile drop-casting, which deserved a great bright application prospect in the marine protective field.

2. Experiment

2.1. Materials and chemical reagents

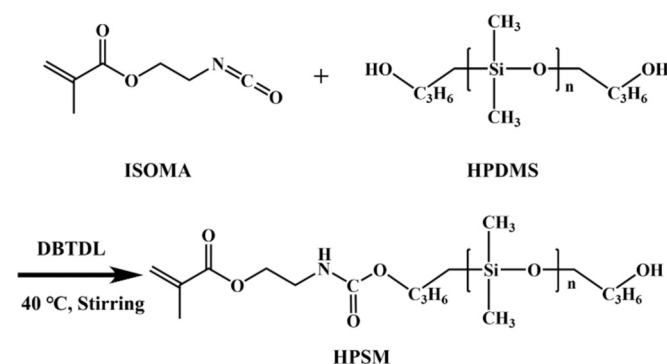
2-Isocyanatoethyl methacrylate (ISOMA, AR grade) was purchased from TCI Co., Ltd. Dibutyltin dilaurate (DBTDL, AR grade) and carbinol (hydroxyl)-terminated polydimethylsiloxane (HPDMS, Mw = 2500) were obtained from Meryer (Shanghai) Chemical Technology Co., Ltd. 1-Methoxy-2-propyl acetate (PMA, GR grade) was purchased from Alighting Reagent (Shanghai), Co., Ltd. Xylene and ethylene glycol methyl ether (EGME, AR grade) were purchased from Sichuan Xilong Science Co., Ltd. Azodiisobutyronitrile (AIBN, AR grade), sodium dodecyl sulfate (SDS, AR grade), benzoic acid and acrylic acid were obtained from Tianjin Kermel Chemical Reagent Co., Ltd. 3-[Dimethyl-2-(2-methylprop-2-enyloxy)ethyl]azaniumyl]propane-1-sulfonate (SBMA, AR grade) and isobornyl methacrylate (IBOMA, AR grade) were purchased from J&K Chemical Ltd. Methyl methacrylate (MMA, AR grade), n-butyl acrylate (BA, AR grade), ethyl acrylate (EA, AR grade) and styrene (St, AR grade) were purchased from Rionlon Bohua (Tianjin) Pharmaceutical Chemical Co., Ltd. 1-dodecanethiol was purchased from Chengdu Kelong Chemical Co., Ltd. A small molecule containing Zn (SMZn) was prepared by acid-base neutralization reaction. Bovine serum albumin (BSA, AR grade) was obtained from Shanghai Macklin Biochemical Co., Ltd. The Micro BCA™ Protein Assay Kit 23,235 was purchased from Thermo Fisher Scientific. Powdered PBS (pH = 7.4) was purchased from Shanghai McLean Biochemical Technology Co., Ltd. *Porphyridium* sp., *Navicula* sp., and their respective culture medium Kock and Erdschreiber were provided by Freshwater Algae Culture Collection at the Institute of Hydrobiology. All the chemicals were used as them was accepted. Deionized water was prepared in the lab using a pure infinity system.

2.2. Preparation of polymerizable organosilicon (HPSM)

The preparation of HPSM is shown in Scheme 1. Firstly, the raw material HPDMS was heated at 120 °C in a vacuum oven for 4 h to remove trace dissolved water completely. Then, the 125 g pretreated HPDMS was transferred into a 250 mL round-bottom flask, heated to 40 °C and stirred with a magnetic stirrer. Afterwards, 0.2 mL of DBTDL catalyst was added into the round bottom flask and stirred evenly. Finally, equal molar ISOMA (7.75 g) was dropped evenly for 3 h. The reaction continued for 4 h after dropping and obtained the polymerizable monomer (HPSM). It is important to note that a strict anhydrous environment must be maintained throughout the reaction process to avoid another side reaction.

2.3. Preparation of smooth liquid-like surface (SLLS)

Firstly, a facile free radical polymerization was applied to prepare



Scheme 1. The diagram of HPSM preparation and the molecular formulas of related reactants.

the functional polymers for a smooth liquid-like surface (SLLS). In this research, polymers with different mass contents (0, 10 %, 20 %, 30 %) of HPSM were synthesized to explore the impact of silicon content on the physical chemistry properties of coatings, and the formulations of polymers were provided in **ST 1**. The polymer with 10 % silicone content was prepared as follows: 10 g EGME, 20 g xylene, 5 g BA, and 6 g HPSM were mixed in a 250 mL three-neck flask, heated to 90 °C, and the mixture was stirred with a magnetic stirrer. At the same time, 3 g SBMA was dissolved with mixed solvent of 15 g EGME and 15 g PMA in a 250 mL beaker, and 75 g SMZn, 10 g EA, 20 g MMA, 6 g BA, 6 g HPSM, 10 g IBOMA, 4 g St and 3 g initiator AIBN were added successively. More notably, 1.5 g chain transfer agent 1-dodecanethiol was provided to adjust the molecular weight of the polymer. When all the reactants were fully dissolved and mixed into a uniform yellow liquid, they were dropped into the above flask for 3 h and then heated for 2 h. Finally, 0.2 g AIBN (dissolved in 5 g EGME) was supplied, and the reaction was continued for 3 h to obtain a polymer with an HPSM content of 10 %. The other polymers were prepared using the same procedure according to **ST 1**.

The SLLS was fabricated using a drop-casting method. More specifically, after the slides were cleaned by ultrasonication with ethanol and acetone, 2 mL polymer diluted with twice the volume of solvent was added dropwise to the slides to cover the whole surface, and when the solvent was evaporated through heating for 24 h at 40 °C, and the coatings were prepared. For convenience in later descriptions, these coatings were marked Free-HPSM, 10 %-HPSM, 20 %-HPSM, and 30 %-HPSM. Unless otherwise specified, the total samples for all characterizations were prepared in this manuscript according to the above method.

3. Characterization

3.1. Characterization of the physical and chemical properties of SLLS

The synthesis of HPSM and the surface composition of SLLS were inspected by Fourier transform infrared (FT-IR) spectroscopy (Nicolet iS10, Thermo Scientific, USA), and the spectra were monitored in the 4000 to 400 cm^{-1} range. The molecular weight and distribution of polymers with different HPSM were determined by gel permeation chromatography (GPC, waters1525 & Agilent PL-GPC220). The glass-transition temperature (T_g) and the thermal properties of SLLS samples were detected by differential scanning calorimetry (DSC) and thermogravimetry (TG) (model: STA 449F3, NETZSCH-Gerätebau GmbH) with a heating rate of 10 K/min, respectively. The chemical compositions of the SLLS were characterized by FT-IR and X-ray photoelectron spectroscopy (XPS, Thermo Scientific, ESCALAB 250Xi) using Al K α radiation as an excitation source. The surface morphologies were inspected by scanning electron microscopy (FE-SEM, JSM-6701F) at 10 kV, the surface roughness was evaluated by a noncontact three-dimensional profilometer ($\times 10$), and root mean square heights (S_q) were accepted as the apparent roughness. The wettability and surface energy were evaluated according to Owen's method [42], where diiodomethane and water were accepted as the probe liquids. Their contact angles were measured by a DSA-100 optical contact angle meter (Kruss Company, Ltd., Germany), in which the static contact angles were measured randomly three times for the samples, and the volume of tested liquid was 5 μL . The coating's mechanical properties (hardness and elasticity modulus) were characterized by nanoindentation (Bruker Hysitron TI980) with continuous stiffness measurement, and a diamond Berkovich indenter was applied. The constant strain rate was 0.1/s, and the maximum load was 10,000 μN . The load was first loaded when it reached the maximum value, kept for 2 s, and finally unloaded for 5 s. The oscillation action was applied when the force was uploaded, its frequency was 220 Hz, and the amplitude was approximately 1–2 nm. Three duplicates were set for each sample.

3.2. Characterization of lubricating property of SLLS

To evaluate the lubricating property of the SLLS, a friction test was conducted on a conventional ball-on-disk reciprocating tribometer (CSM Co., Ltd., Switzerland) by acquiring the friction coefficient (μ) at different given load and frequency. When it was tested, water was used as the lubricant medium. The corresponding load, amplitude and frequency were 1 N, 0.5 mm, and 1 Hz, respectively, and the friction counterpart was a polypropylene ball with a diameter of 6 mm.

3.3. Characterization of protein adsorption

To evaluate the protein adsorption behavior of SLLS, BSA was chosen as a nonspecific model to evaluate the anti-adsorption capability quantitatively. The specific protocol of the test was as follows. The sample was cut into small pieces with an area of $1 \times 1 \text{ cm}^2$ and soaked in PBS solution for 24 h to reach swelling equilibrium. Then, the pieces were dried with N_2 , placed in another 24-well culture plate and immersed in 2 mL BSA solution (0.5 mg/mL, PBS solution as solvent). The samples were incubated at 37 °C for 4 h to adsorb protein fully and then rinsed three times with PBS solution to remove non-adhered protein. The adsorbed protein on the SLLS was detached with 1 wt% SDS (2 mL, deionized water as solvent) at a shaking speed of 180 rpm at 37 °C for 60 mins. Finally, the concentration of adsorbed protein was determined by the Micro-BCA kit, and the amount of the adsorbed protein was calculated from the UV absorption standard curve (**Fig. S1**) at the peak of 562 nm. The standard curve was derived from the standard BSA solution (2 mg/mL). Each measurement was performed in triplicates, and the average value was accepted.

3.4. Characterization of antifouling property of SLLS

The antifouling property of SLLS was determined using two kinds of marine algae cultured in the lab. The details of the protocols of this test were shown in our previous work [43]. In brief, when microorganisms fouled the samples after 24 h incubation, the numbers of adhered algae were determined using the same method in the literature [15].

Furthermore, the marine field test was conducted to inspect antifouling performance in harsh marine environment. The tests were done for eight cycles, each cycle consisted of a dynamic exposure for 30 days and static exposure for 10 days according to GB-T 7789–2007.

3.5. Characterization of drag reduction behavior of SLLS

The drag reduction behavior of SLLS was measured by a Couette flow field constructed by a rotary rheometer (**Fig. S2**). For experiments, the SLLS was coated on glass discs with a diameter of 35 mm, the same as the rotor of the rheometer. The samples were fixed on the support platform of the rheometer, and to avoid the disorder of the flow field caused by water evaporation, the interval (1 mm) between the upper plate and the sample was filled with a mixture of glycerol and water (mass ratio = 2:1). Meanwhile, to visually observe the velocity variation diagram of the flow field, a $\mu\text{-PIV}$ (Micro-Particle Image Velocimetry, LaVision GmbH, Anna-Vandenhoeck-Ring 19, D-37081 Göttingen) was used to characterize the flow field. The interval was 0.5 mm, the shear rate was 0.5 s^{-1} , and proportional fluorescent tracer particles of 2 μm were added to the working fluid to obtain the velocity profile of the flow field. Two frame modes were adopted for shooting; the first frame's exposure time was 2000 μs , and the interval between the two frames was 500 μs . The size of the inquiry window was 64 pixels $\times$ 64 pixels, and the overlap area was 50 %. One hundred images were taken in 7.04 s.

4. Results and discussion

4.1. The preparation of HPSM and functional polymers

Firstly, to integrate the silicone monomer (HPDMS) into the polymer chain, the polymerizable monomer (HPSM) was prepared, and FT-IR characterized the preparation process. As shown in Fig. 1a, the modified infrared spectrum showed that the characteristic absorption peaks of -NH- (1530 cm^{-1}), C=C (1640 cm^{-1}), C=C-H (3050 cm^{-1}) and C=O (1730 cm^{-1}) were presented, which proved that HPSM was successfully prepared. Then, a variety of polymers with a different mass content of HPSM were synthesized, and the formulation was shown in Fig. 2. Various allyl monomers were integrated into the random polymer to achieve good comprehensive properties, such as compliance, thermal properties, and mechanical stability. For example, the monomers MMA, BA and EA could improve the thermal properties and adjust the glass transition temperature of the copolymer. In particular, IBOMA was used to reduce the internal viscosity, while SMZn was introduced to integrate the polymer with self-renewal ability. More importantly, HPSM was employed to adjust the physical and chemical properties of the coating formed with the polymer. The polymerization of these monomers was characterized by FT-IR and XPS spectra. There were absorption peaks (at 1100 cm^{-1} and 810 cm^{-1}) of the polymer with silicon monomers, which were attributed to the stretching and bending vibration of the Si-O bond, and the peak at 1260 cm^{-1} was ascribed to the absorption of the C-O bond of the ester group. Meanwhile, the characteristic absorptions of 1750 cm^{-1} (carbonyl), 2960 cm^{-1} (-CH_3), 2870 cm^{-1} ($\text{-CH}_2\text{-}$), and 3450 cm^{-1} (-NH-) were enhanced, which indicated that the above monomers with specific properties were successfully introduced into the polymer chain (Fig. 1b). The molecular weight distribution and thermal properties of these polymers were also valued in Table S2. The polydispersity indexes (PDIs) of the prepared polymers were about 2.

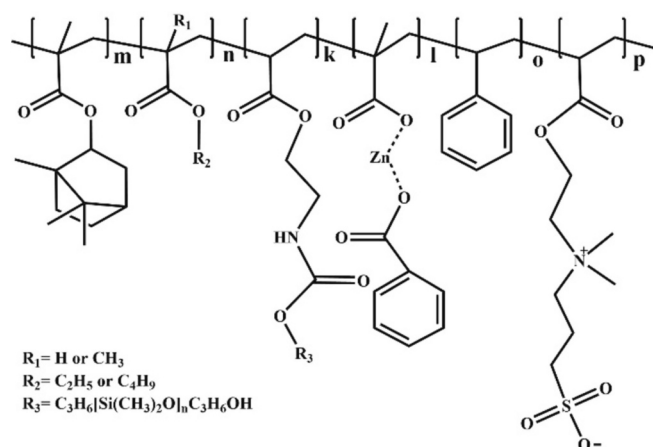


Fig. 2. The formulation of random polymer.

Thermal analysis implied that the polymers had a suitable glass transition temperature (approximately $10\text{--}20\text{ }^\circ\text{C}$) to prepare a functional coating. Meanwhile, the organosilicon (HPSM) can effectively improve its thermal decomposition temperature, changing from $80\text{ }^\circ\text{C}$ to $195\text{ }^\circ\text{C}$, and with the increase of HPSM, the remaining mass of the polymer also increased from 20 % to 30 %, as shown in Fig. S3. In addition, when the coating was prepared, XPS also analyzed the surface composition of the polymer film. The results indicated that the mass ratio of Si/C and silicon mass continuously increased with the HPSM. (Table S3 and Fig. 1c-d). The higher silicon content indicated that the surface was more prone to surface reconstruction and favored smooth surface and special wettability [44]. This state was precisely like a layer of liquid covering the surface, which can adjust the conformation of surface molecules in

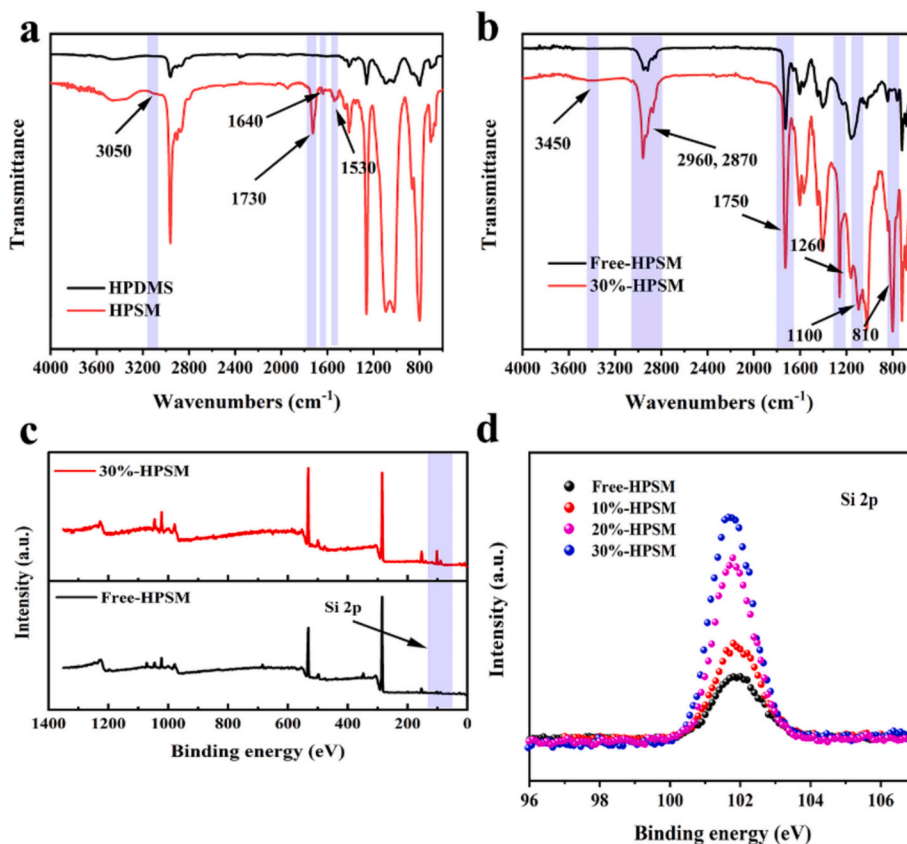


Fig. 1. Analysis of the chemical composition of the prepared HPSM and polymers. (a) The FT-IR spectrum of HPSM and serial polymers. (b) The XPS full spectrum (c) and fine spectrum of characteristic elements Si 2p (d) when the polymer was coated, respectively.

time when something comes into contact with the surface. Therefore, the coating was called a smooth liquid-like surface (SLLS).

4.2. Surface morphology of SLLS

The unique chemical composition and particular structure of materials dominate in determining their performances and application. Therefore, when the surface chemical composition of SLLS was inspected, their surface structure and section morphology features were examined. As shown in Fig. 3, the SLLS surfaces were smooth and homogeneous, and the cross-sectional structure was uniform and dense, without any apparent defects. Obviously, the introduction of HPSM monomers did not cause phase separation or component segregation, which resulted from poor compatibility between different components and always promoted the loss of the specific function of the coating [45]. In addition, the roughness of the SLLS was extremely small, reduced from 100 nm to 49 nm, which indicated that the HPSM occupied good compatibility with other monomers. The excellent compatibility imparted the long organic silicon molecular chains with a significant ability to move freely. It was conducive to their enrichment on the coating surface, which agreed with the results in Fig. 1.

4.3. Characterization of the physical and chemical properties of SLLS

For functional coatings, surface wettability was important for determining the adsorption and desorption properties when other substances contacted the surfaces, which is also suitable for marine antifouling coatings. Generally, the wettability of a solid surface is determined by both the chemical composition and surface structure. Therefore, when the characterization of the composition and surface morphology of SLLS were conducted, the surface wettability was measured by the contact angles of deionized water and diiodomethane

according to the Owens-Wendt (OW) method [42]. As shown in Fig. 4a, with the increase of HPSM content in the coating, the contact angle of water increased from 72° to 92°. Noticeably, the surface energy of the coating changed from 34 mJ/m² to 22 mJ/m², and the smaller value of 22 mJ/m² was precisely equal to the dispersive component of the surface energy of water and significantly reached the minimum on the Baier curve which depicted the relationship between relative adhesion (dimensionless) of marine fouling and the surface tension of polymer materials [46]. These results implied that augmenting the content of HPSM can effectively weaken the adhesion action of marine organisms. On the other hand, although the resistance to fouling was influenced by multiple factors, such as bulk property, thickness and surface action (chemical bond, mechanical interlocking and diffusion), both the critical surface energy (γ_c) and the elastic modulus (E) of the polymer coating were most frequently correlated with bioadhesion. According to the theory of fracture mechanics, the critical removal force is proportional to the value of $\sqrt{(E \times \gamma_c)}$ [46]. Therefore, the mechanical properties of the coating, including hardness and elastic modulus, were evaluated by nanoindentation test. As the curves depicted in Fig. 4b, the creep step of polymer coatings broadened with the increase of HPSM content in the maximum load hold time, which was associated with the viscoelasticity of the polymer. In detail, the viscous flow of molecular chain were restrained in the loading process and fully developed in the load retaining stage, showing obvious creep deformation at the maximum load. Compared with the control (Free-HPSM), the presence of the HPSM augmented the freedom of polymer main chain movement, which contributed to the conformation transition of polymer chain on the surface and the good flexibility of surface. This mechanism made it difficult for fouling organisms to stick strongly to the SLLS. Obviously, the hardness and elastic modulus of the SLLS were also significantly reduced due to the addition of HPSM, the corresponding hardness was reduced from 200 MPa to 60 MPa, and the elastic modulus was changed

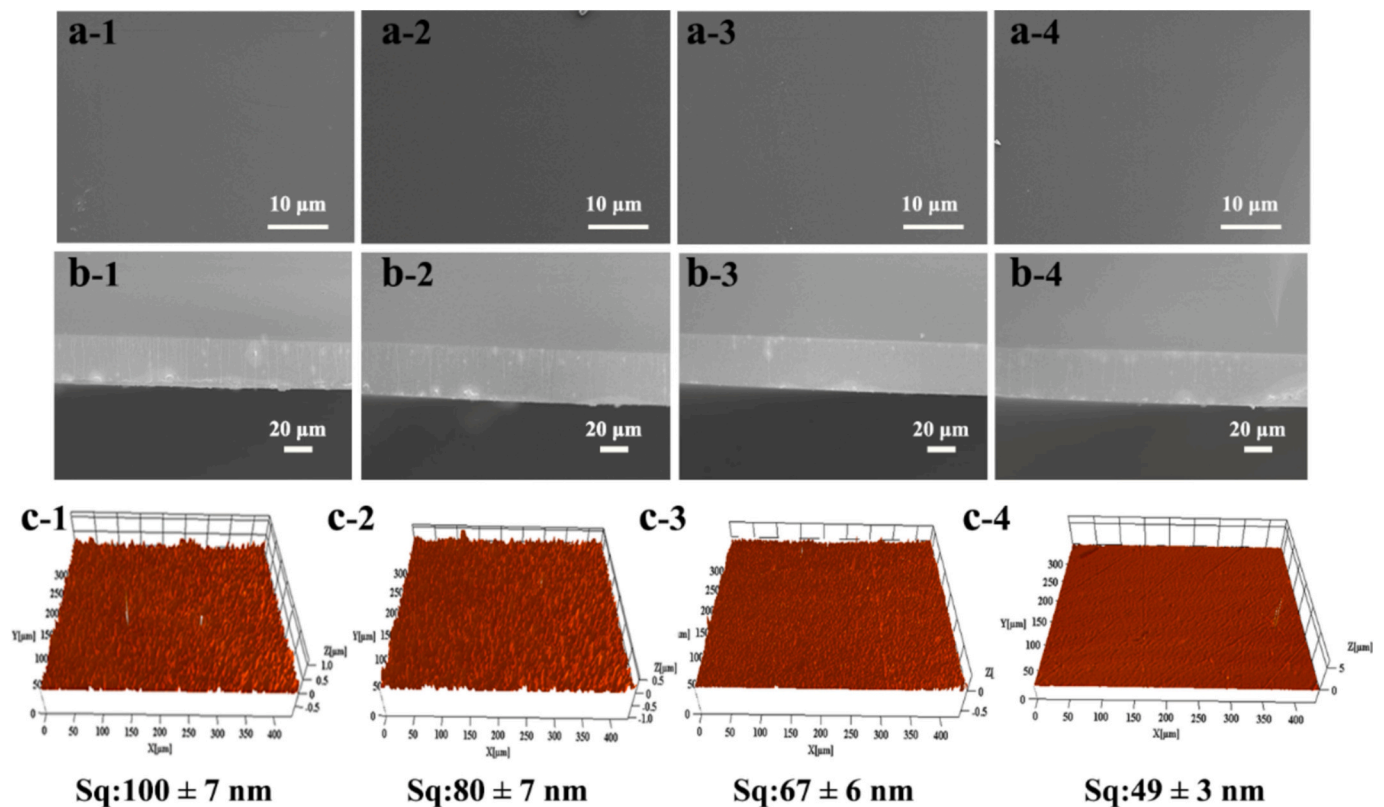


Fig. 3. Characterization of surface morphology and roughness of smooth liquid-like coatings. SEM images of the surface (a), cross-section (b) and (c) typical three-dimensional contour height images of the coating surfaces for (1) Free-HPSM, (2) 10 %-HPSM, (3) 20 %-HPSM, (4) 30 %-HPSM and their corresponding roughness, respectively.

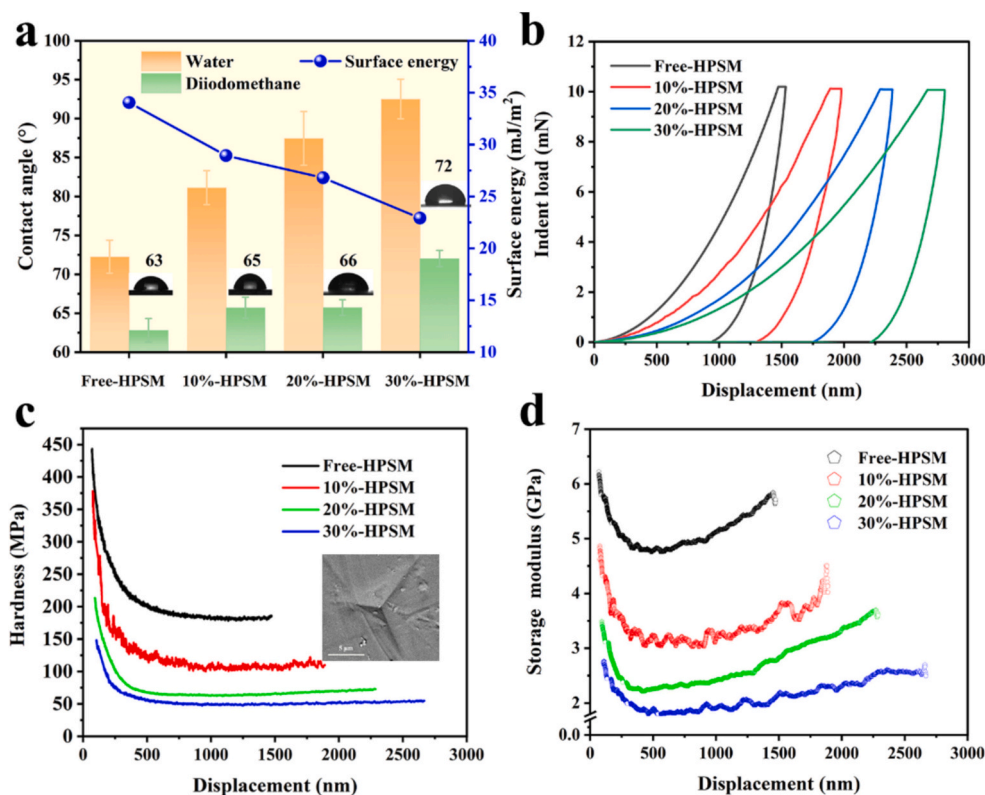


Fig. 4. Characterization of the surface wettability and mechanical properties of SLLS. (a) Contact angles of polar (purified water), nonpolar (diiodomethane) liquids and surface energy of various coatings. (b) Load versus displacement curve during the nanoindentation test and their corresponding (c) hardness and (d) elastic modulus curves. The inset in the lower right corner of picture (c) showed the trace left by the indenter during the test.

from 4.8 GPa to 1.8 GPa (Fig. 4c-d). Combining the above discussion of the surface tension, it was reasonable to deem that HPSM monomer can effectively weak the adhesion of biofouling on the polymer coating, which was attributed to special viscoelasticity and surface flexibility. Moreover, the HPSM did not cause a compromise of coating adhesion according to Fig. S4, and the adhesion grade was all above 4B, surpassing the the basic performance requirements for common coating.

4.4. The resistance to protein adsorption of SLLS

The proteins in the marine environment will adsorb on the fresh surface to form a condition film within a few seconds or minutes when the solid surface is immersed, which will provide a supporter for the nutrient exchange of numerous bacteria, algae and barnacles spores. Subsequently, a great deal of microorganisms (bacteria, algae and others) was attached to the surface, and an irreversible macro biofouling was developed [47,48]. Therefore, strong resistance to protein was necessary for their comprehensive antifouling properties. The typical bovine serum protein (BSA) was selected as a model protein to evaluate the anti-protein adsorption ability of the prepared SLLS (Fig. 5). The amount of protein adsorption was $5.3 \mu\text{g}/\text{cm}^2$ for Free-HPSM, and the amount was descended with the increase of HPSM. When the HPSM mass content was 30 %, it was reduced to $2 \mu\text{g}/\text{cm}^2$. The results showed that SLLS possessed strong resistance to BSA protein, the excellent anti-protein capability of SLLS was attributed to its unique surface properties. It was inferred from Fig. 4a that the surface became more hydrophobic than Free-HPSM and the surface tension had declined with the increase of HPSM. This tendency is not conducive to hydrophilic protein adhesion. The test indicated that the SLLS has strong anti-protein adsorption capability.

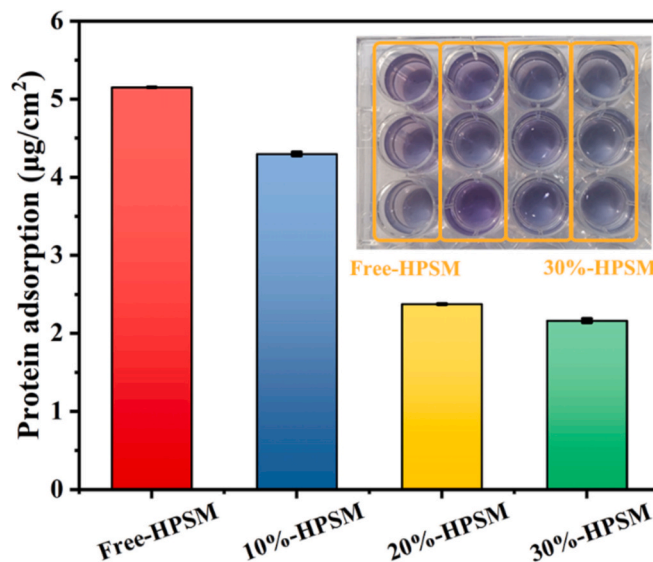


Fig. 5. The resistance of representative protein adsorption. The histogram of statistical protein adsorption mass according to the UV absorbance in 562 nm, the inset was the photograph of detached protein dyed with the Micro-BCA kit. The deeper colour represent the mucher the content of protein. Error bars represented the standard deviation of three duplicates.

4.5. The assessment of antifouling behavior of SLLS

Two kinds of representative fouling algae were chosen to evaluate the resistance of the SLLS surface to the alga biofouling. The SLLS with HPSM had a strong ability to inhibit the adhesion of both red algae

(*Porphyridium* sp.) and diatom algae (*Navicula* sp.) after 24 h incubation (Fig. 6a-d, a'-d'). In detail, for *Porphyridium* sp., the adhesion density decreased from 2990 /mm² to 1700 /mm², 700 /mm², and 480 /mm² in succession, and the corresponding antifouling ratio changed from 43 % (10 %-HPSM) to 83 % (30 %-HPSM); for *Navicula* sp., the adhesion density in turn changed from 1480 /mm² to 870 /mm², 410 /mm² and 360 /mm², and the corresponding antifouling ratio ascended from 41 % to 76 %, respectively (Fig. 6e-f). These results manifested that the integrated organosilicon endowed the random copolymer with strong inhibition of marine algae adhesion. The excellent antifouling performance of SLLS was ascribed to two main factors. One can be explained that a freely-rotating silicon-oxygen chain promoted great freedom of motion of functional groups on the polymer. This freedom of motion of functional groups restrained their ability to close to, orient with, and form dipolar or hydrogen bonds with complementary functional groups on marine adhesives [46]. The other character of interface renewal owing to the self-polishing component made it more difficult for algae to adhere to the coating surface, further enhancing the antifouling performance of the SLLS. This behavior of prohibition algae fouling agreed with the discussion of the above anti-protein adsorption.

4.6. The assessment of drag reduction behavior of SLLS

Marine fouling will inevitably result in an increase in hull navigation resistance, which will adversely impact marine transport [49,50]. The organosilicon was a competitive candidate with strong anti-fouling and drag-reduction capabilities due to its unique viscoelastic properties and low surface energy [51,52]. In this research, when the excellent anti-fouling properties of the SLLS were guaranteed, the coating's solid-solid and solid-liquid friction resistance was evaluated by friction experiment and plate shear flow, respectively. For the friction test, the whole friction curves were smooth during the test process, and the friction coefficient ranged from 0.135 to 0.06 with the increase of HPSM in Fig. 7a-b. The results manifested that the friction resistances between SLLS and friction pair decreased significantly. Friction resistance was closely related to the physical and chemical properties of contact surfaces, such as the roughness and surface composition, due to the partial direct contact between the friction pair and SLLS. According to Fig. 1, the content of Si element on the coating surface was positively related to the HPSM, while the compound with silicon can impart material with good lubrication performance [53,54]. Besides, the extremely low roughness and smooth surface structure of SLLS (in Fig. 3) effectively avoided the unsmooth stagnation of the contact area. Therefore, it was not difficult to conclude that the SLLS surface was smooth enough to have excellent self-

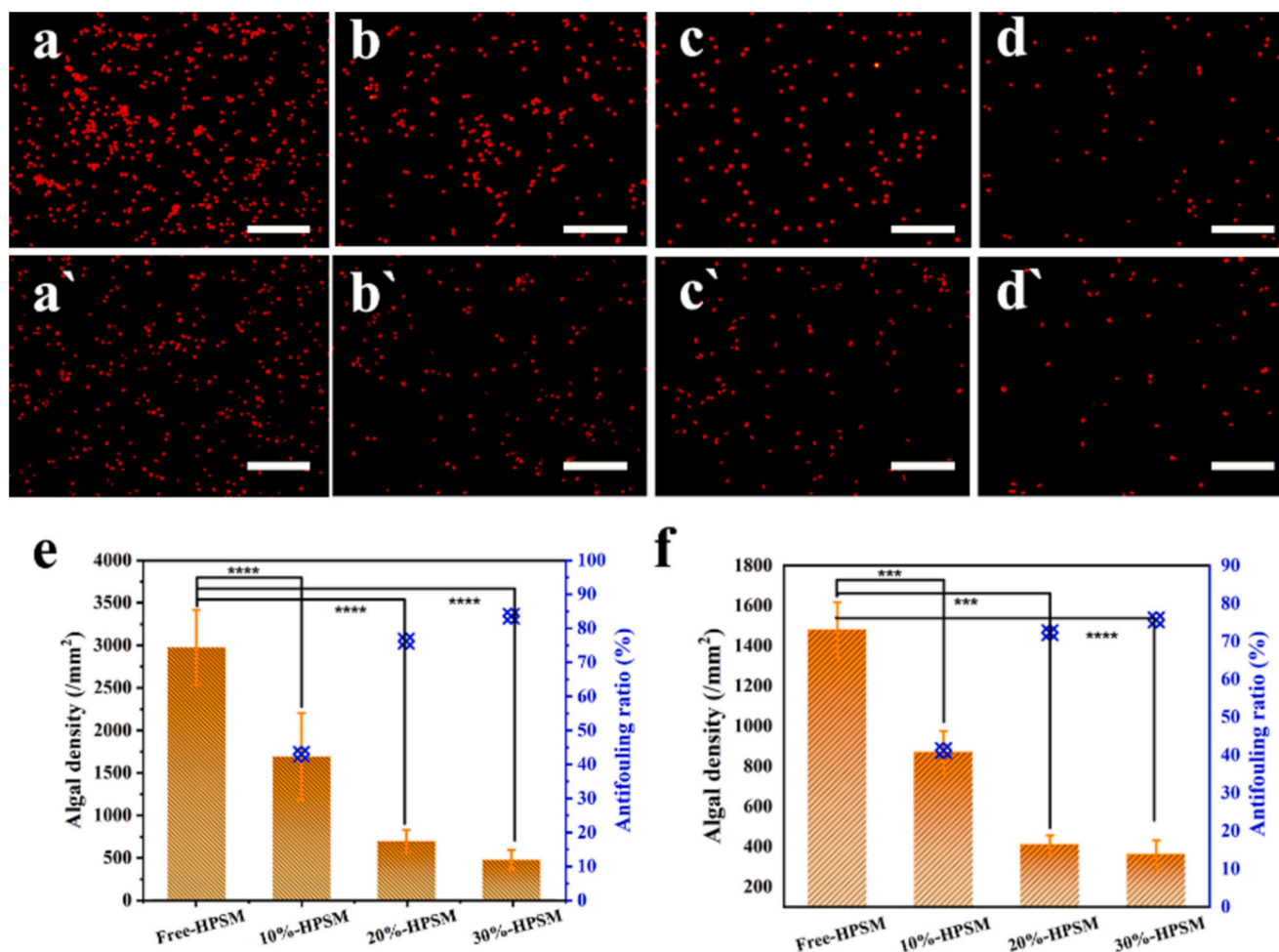


Fig. 6. The assessment of the antifouling performance of SLLS employing two typical marine algae. The representative fluorescent microscope images of algae attached to the SLLS incubated at 25 °C for 24 h. a, b, c, d (*Porphyridium* sp.) and a', b', c', d' (*Navicula* sp.) corresponded with Free-HPSM, 10 %-HPSM, 20 %-HPSM, 30 %-HPSM, respectively. The graph e (*Porphyridium* sp.) and f (*Navicula* sp.) were the relevant statistical adhesive algae densities and their antifouling ratios based on the adhesive number of Free-HPSM, where the antifouling ratio was defined as $(n_c - n_s) / n_c \times 100 \%$, where n_c and n_s respectively were the numbers of adhered algae of control (Free-HPSM) and test samples (SLLS with HPSM). (p values: $0.01 < *p < 0.05$, $0.001 < **p < 0.01$, $0.0001 < ***p < 0.001$, $****p < 0.0001$ all the values are expressed as mean $\pm$ SD, $n = 10$)

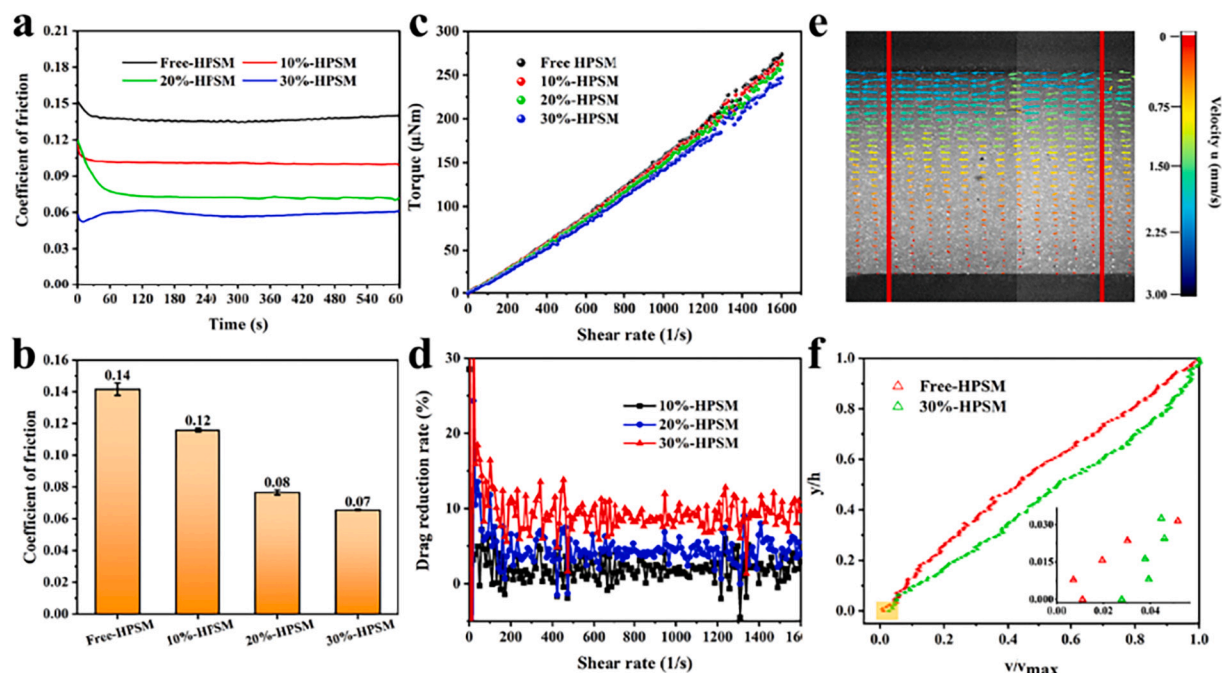


Fig. 7. The assessment of drag reduction performance of SLLS. (a) and (b) were frictional curves and corresponding friction coefficients for SLLS by friction test. (c) The torque versus shear rate (flow velocity) curve and drag reduction rates (d) of SLLS at different speeds. (e) Velocity vector fields between two coaxial circular plates superimposed with PIV images and (f) normalized velocity profiles of flow on SLLS according to velocity vector fields.

lubricating performance, effectively reducing the friction of solid surfaces in contact.

For the navigation body, the small friction resistance of the solid-liquid interface was more critical than the solid-solid friction resistance. Afterwards, the solid-liquid coupling effect was tested with a rotational rheometer, and the drag reduction rate (DR) was calculated based on the torque force. The drag reduction rate was defined as:

$$DR = \frac{\tau_c - \tau_s}{\tau_c} \times 100\%$$

Hereon, τ_c and τ_s respectively were the torque force of control and test samples under the same test condition. The sample Free-HPSM was accepted as a control.

As Fig. 7c-d depicted, the SLLS with HPSM showed excellent drag reduction performance in the test, and its DR was 3 %, 5 % and 10 % in sequence. Moreover, the flow field distribution on the SLLS was captured by PIV investigation, and their velocity profiles were obtained in Fig. 7e-f. Notably, at the same height (y), the sample of 30 %-HPSM had a higher velocity than the Free-HPSM in Fig. 7f, which implied the flow field had a low-velocity gradient in the near wall. The small velocity gradient was the indication of higher DR. The velocity slips of the wall were present, and the slip lengths were varied from 5 μm to 100 μm (Fig. S5). Different from the drag reduction mechanism of hydrophilic coatings [55], the function of drag reduction of SLLS was attributed to the deformation caused by its special viscoelasticity when water flows on the surface, called spanwise transversal [56]. This elastic deformation can promote drag reduction efficiency in two aspects. On the one hand, this wavy deformation effectively absorbed the kinetic energy of flow and suppressed the pressure pulsation of the fluid. On the other hand, due to the impact of water flow, hydrophobic and flexible polymer chains may form a rippled surface. While the rippled surface was a gas-liquid-free shear surface, effectively increasing the thickness of the boundary layer [57]. The above two effects caused the transition delay from the laminar boundary layer [58,59]. However, when the wave's amplitude exceeded the thickness of the boundary layer in the near-wall region, the continuous boundary layer was interrupted, which increased the strength and resistance of the disordered flow. Therefore, the

appropriate deformation was necessary for the specific flow field to reduce the resistance. Furthermore, the smooth surface of SLLS also played an essential role in determining the drag reduction efficiency. Specifically, because the surface was smooth enough without any additional protrusions, the viscous bottom layer of the liquid was intact, and the wall did not increase the resistance. The hydrophobic silicone molecular chains, like a layer of freely moving liquid film attached to the coating surface, endowed the surface with special viscoelasticity, wettability and smoothness and reduced the resistance of the solid-liquid composite interface. From the above discussion, it was reasonable to deem the SLLS with HPSM possessed excellent drag reduction performance of the solid-solid and solid-liquid interface.

When the various properties of SLLS were evaluated in the laboratory, the marine field test was conducted to further assess its antifouling capabilities and overall performance in harsh marine environments. The results were shown in Fig. 8. There was substantial nematode colonization and minor barnacle attachment for the control panel after two testing cycles (Fig. 8a), and a significant accumulation of biofilm subsequently formed, nearly covering the entire surface of the plane (Fig. 8b). Macroscopic biofouling happened after the sixth cycle, a great deal of macroalgae barnacles completely covered of panel surface, this biofouling intensified in the eighth cycle (Fig. 8c-d). In contrast, the SLLS demonstrated remarkable resistance to biofouling throughout the entire testing period. The coating surface remained intact and smooth against exposure of eight cycles. These results manifested that the SLLS possessed excellent antifouling performance and mechanical stability in the complex ocean environment. The outstanding antifouling properties were primarily attributed to the synergistic effects between the unique characteristics of resin, including low surface energy and low elastic modulus, and its self-polishing mechanism. Furthermore, the superior drag-reducing properties further prevent fouling organisms from adhesion. These findings stood for significant potential application of this novel coating in the protection of marine equipment and infrastructure.

5. Conclusions

In this research, a novel organosilicon macromolecule (HPSM) was

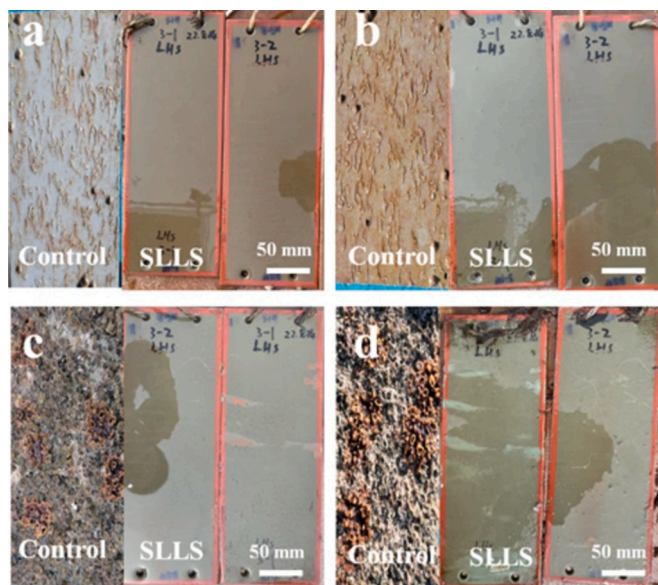


Fig. 8. The marine field test of SLLS and the control plate with epoxy resin for different period. The photographs for (a) two test cycles, (b) four test cycles, (c) six test cycles and (d) eight test cycles. The sample size was 210 mm × 80 mm × 3 mm.

synthesized, and the HPSM was chemically bonded to the self-polishing polymer chains to fabricate novel antifouling and drag reduction coatings. By the hydrophobic and flexible molecular chains of HPSM, the coating was like a layer of liquid and hence was named a smooth liquid-like surface (SLLS). Interestingly, the SLLS with a proper amount of HPSM possessed lots of unique properties, such as wettability, viscoelasticity, surface smoothness and tension. The antifouling and drag reduction performances were assessed. The results indicated that the integration of HPSM imparted the coating with strong resistance to typical protein and marine algae. With the increase of HPSM, this ability was continuously strengthened and ascribed to the unique physico-chemical property (surface energy and elasticity modulus). For 30 %HPSM coating, the surface energy was 22 mJ/m² and near the optimal value for antifouling material to remove adhesion. Deservedly, the strong antifouling performance was attributed to the synergistic effect of the liquid-like behavior and the self-polishing mechanism of the coating. Meanwhile, excellent drag reduction performance was also achieved. This research developed a convenient method to fabricate a novel smooth antifouling and drag reduction coating by a hydrophobic and moving freely organosilicon polymer chain, which can effectively avoid lubricant depletion at the interface of traditional slippery liquid-infused surfaces. Considering the facile preparation, excellent antifouling and drag reduction performance, this research has provided an attractive and alternative material in marine antifouling and drag reduction.

CRedit authorship contribution statement

Jianbin Zhang: Writing – original draft, Investigation, Funding acquisition, Data curation, Conceptualization. **Meijun Feng:** Methodology, Formal analysis, Data curation. **Xiaotong Wu:** Software, Methodology, Investigation, Conceptualization. **Qiangliang Yu:** Resources, Investigation, Funding acquisition, Formal analysis. **Daoli Wang:** Supervision, Project administration, Methodology, Investigation, Funding acquisition. **Libang Feng:** Supervision, Software, Resources, Methodology, Funding acquisition, Conceptualization. **Yanhua Liu:** Supervision, Software, Resources, Investigation, Formal analysis. **Wufang Yang:** Writing – review & editing, Supervision, Funding acquisition. **Xiaowei Pei:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Feng Zhou:** Supervision, Project

administration, Funding acquisition.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper. The authors have no relevant financial or non-financial interests to disclose. All authors have approved the manuscript and agree with its submission to *Surface and Coatings Technology*.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.surfcoat.2024.131547>.

Data availability

No data was used for the research described in the article.

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